

Instruction Pipelining, Hazards & Cache Memory Organization

Subject: CS403 | Unit 3: Processor Design & Memory

1. Executive Overview & Syllabus Scope

Covers 5-stage MIPS pipeline IF, ID, EX, MEM, WB, Structural/Data/Control hazards, Forwarding, Branch prediction, Direct/Set

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 3: Computer Architecture & Pipelining Organization
- 1. 5-Stage RISC Pipeline: Instruction Fetch IF, Instruction Decode ID, Execute EX, Memory MEM, Writeback WB
- 2. Pipeline Speedup Formula: Ideal speedup $S = k$ stages under $CPI = 1$ assumption
- 3. Pipeline Hazards: Structural hazards, Data hazards RAW, WAR, WAW, Control hazards Branches
- 4. Hazard Resolution Techniques: Stalling bubbles, Data Forwarding/Bypassing, Branch Prediction
- 5. Cache Mapping Techniques: Direct-Mapped, Fully Associative, and N-Way Set Associative
- 6. Cache Performance: Hit Ratio, Miss Penalty, and Average Memory Access Time AMAT

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.